



12274 - The Faint Frontier: Pushing Proper Motion Limits with JWST+Gaia

Cycle: 5, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Kevin Andrew McKinnon (PI) (CSA Member)	University of Toronto
Dr. Roeland P. van der Marel (CoI) (US Admin CoI)	Space Telescope Science Institute
Liwen Chen (CoI) (CSA Member)	University of Toronto
Dr. Mattia Libralato (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Padova
Dr. Andres del Pino Molina (CoI) (ESA Member)	Instituto de Astrofisica de Andalucia (IAA-CSIC)
Dr. Jay Anderson (CoI)	Space Telescope Science Institute
Dr. Sangmo Tony Sohn (CoI)	Space Telescope Science Institute
Dr. Paul Bennet (CoI)	Space Telescope Science Institute
Jack Thomas Warfield (CoI)	The University of Virginia
Dr. Laura L. Watkins (CoI) (ESA Member)	Space Telescope Science Institute - ESA - JWST
Dr. Eduardo Vitral (CoI) (ESA Member)	University of Edinburgh, Institute for Astronomy
Dr. Mark Fardal (CoI)	Eureka Scientific Inc.
Dr. Ivanna Escala (CoI)	Space Telescope Science Institute
Aviad Levis (CoI) (CSA Member)	University of Toronto
Prof. Gwendolyn Marie Eadie (CoI) (CSA Member)	University of Toronto
Dr. Andrea Bellini (CoI)	Space Telescope Science Institute
Dr. Ting Li (CoI) (CSA Member)	University of Toronto
Dr. Massimo Griggio (CoI)	Space Telescope Science Institute
Dr. Nitya Kallivayalil (CoI)	The University of Virginia
Dr. Andrew Pace (CoI)	The University of Virginia

OBSERVATIONS

ABSTRACT

We propose to develop an open-source tool that will combine JWST images with Gaia data to measure significantly-improved stellar astrometry. Leveraging the high resolution of JWST images and their ever-increasing time baseline to Gaia means that JWST+Gaia proper motions (PMs) will be tens of times more precise than Gaia alone, especially for faint stars ($G > 18$ mag). For the faintest stars with only Gaia-measured positions ($G > 20.7$ mag), we will report first-ever PMs.

This new tool, dubbed GaiaWebb, will automatically download and process NIRCам, NIRISS, and MIRI images, cross-match JWST sources with Gaia stars, align the JWST images to the Gaia reference frame, and return more precise stellar motions than either observatory can achieve on its own. With the completed GaiaWebb pipeline, we will release catalogs of improved PMs for stars in archival JWST images. The GaiaWebb tool and its associated astrometric catalogs will benefit the whole astronomical community, especially for studies of Local Group kinematics and dynamics. In particular, high precision PMs of faint stars will enable detection and characterization of new substructure in the Milky Way (MW), which will constrain models of dark matter (DM) and galactic formation and evolution.

OBSERVING DESCRIPTION